TEXORA · LECTURE SHEET

Warp Knitting

Every needle fed by its own warp yarn, all loops of a course formed in one instant, and the structure built by the sideways shifts — shogs — of guide bars between courses. Warp knitting is the high-speed, run-resistant half of the knitting industry: tricot for lingerie and linings, raschel for lace, nets and technical textiles.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 13 · LEVEL 3 TERM 1

1. Principle and machine elements

Yarn comes from **warp beams** (one per guide bar), each beam carrying one yarn for every needle across the working width. **Guide bars** — bars carrying individual yarn guides (eyes) — swing the yarns around the needles (**overlap**: the swing across the hook side that lays yarn into the hooks) and shift sideways along the needle bar (**underlap**: the movement behind the needles that carries yarn from one wale to another). Needles (latch or compound), sinkers and a presser/knock-over arrangement execute the same loop cycle as weft knitting, but for all needles simultaneously. A **pattern mechanism** (pattern chain/wheels, or electronic guide-bar control) programmes the shog sequence. Two machine families: **tricot** — fine gauge, bearded or compound needles, usually 2–4 guide bars, high speed, plain and simple patterned fabrics; **raschel** — latch needles, coarser-to-medium gauges, many guide bars (pattern bars plus ground bars), the lace and technical machine.

2. The loop-forming cycle (tricot, latch/compound form)

1. **Swing-in / overlap:** guide bars swing toward the needles and shog sideways (usually 1 needle space) so each guide lays its yarn into the path of its needle's hook.
2. **Pressing/closing and knock-over:** needles rise to clear, guides swing back, needles descend drawing the new yarn through the old loops, which knock over — a complete course exists.
3. **Underlap shog:** behind the needles, the guide bars shift by the number of needle spaces the structure demands (0 for pillar stitch, 1–2 for tricot, more for atlas/satin effects). This shift decides which wale the next course's loop will connect to — it *is* the fabric design.

Because all needles act together, course rates are very high (tricot commonly 1000+ courses/min; see §5), but width is fixed by the beam and pattern changes need new shog programmes and beam preparation.

3. Guide-bar lapping and chain notation

A lapping movement is recorded on point paper as the path of one guide around needle spaces, course by course, and numerically as **chain notation**: for each course, the needle-space numbers between which the guide stands before and after the overlap shog, written in course order. The difference within a pair = overlap (normally 1); the change between courses = underlap. Standard ground structures (two guide bars, front bar listed first in this sheet):

STRUCTURE	BAR 1 LAPPING	BAR 2 LAPPING	CHARACTER
Pillar (chain) stitch alone	1-0 / 0-1 repeated, underlap 0	—	Separate chains, no fabric — must combine with inlay or a second bar
Tricot (1 and 1)	1-0 / 1-2	1-2 / 1-0 (opposite)	The standard closed tricot ground: balanced, stable
2 and 1 (satin-like longer underlap)	1-0 / 2-3	2-3 / 1-0 (opposite)	Longer floats/underlaps, smoother face, more lustre
Atlas	Progressive shog each course, then return (e.g. 1-0/1-2/2-1/2-3/3-2/3-4…)	Mirror/opposite	Diagonal/matt-shine effects
Inlay (weft insertion)	Ground bar as tricot	Inlay bar: underlap only, no overlap (never enters hooks)	Yarn laid straight across — stability, stretch control, elastane laying-in

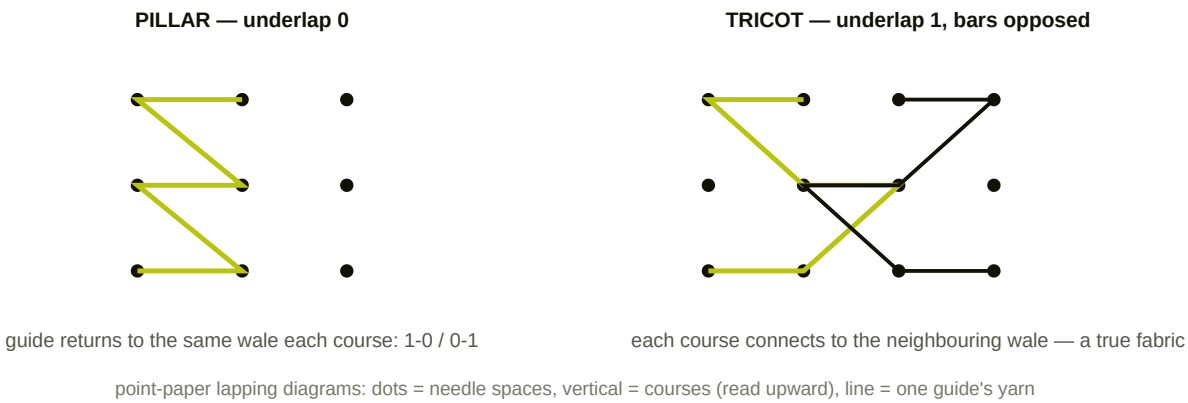


Figure 1 — Lapping diagrams for pillar and two-bar tricot. The underlap — the sideways step between courses — is the difference between separate chains and a connected fabric.

4. Run-in, stitch length and fabric control in warp knitting

Warp stitch length is not set by a stitch cam but by **run-in**: the yarn length fed per **rack** (480 courses — the industry counting unit) for each guide bar, measured in mm/rack. Longer run-in per course = longer loops = looser, heavier-per-loop but lower-CPC fabric. Each bar has its own run-in (front bar usually slightly different from back bar to balance the fabric). Yarn consumption and fabric weight follow from run-in directly: yarn per bar per hour = run-in × courses/hour ÷ 480.

5. Production — worked example (verified)

WORKED EXAMPLE — TRICOT LENGTH, YARN USE AND WEIGHT

Machine: 130 in working width, E28, 2 bars, 1200 courses/min, relaxed CPC = 22/cm. Run-in: Bar 1 = 1520 mm/rack, Bar 2 = 1480 mm/rack. Yarn 8.3 tex (75 denier) both bars, efficiency 90 %.

Fabric length = 1200 × 60 ÷ 22 ÷ 100 = 32.73 m/h theoretical; × 0.90 = **29.5 m/h**.

Courses/hour = 72,000 = 150 racks/h. Run-in is quoted per yarn end. Ends per bar = 130 in × 28 = 3640. Bar 1: per end per hour = 150 racks × 1520 mm = 228,000 mm = 228 m; all ends = 3640 × 228 m = 829,920 m = 829.9 km → × 8.3 g/km = 6.89 kg/h. Bar 2: per end = 150 × 1480 mm = 222 m; total = 3640 × 222 = 808,080 m = 808.1 km → 6.71 kg/h.

Total yarn = **13.60 kg/h** theoretical; × 0.90 = **12.2 kg/h**. Cross-check GSM: 12.2 kg/h ÷ (29.5 m/h × 3.30 m width) = 12,200 ÷ 97.35 = **125 g/m²** — plausible for a two-bar tricot of this yarn. ✓

6. Warp versus weft — closing comparison

FEATURE	WARP KNITTING	WEFT KNITTING
Loops of a course	All formed simultaneously	Formed across the course by feeder/carriage
Yarn supply	Beams, one end per needle per bar	Cones, one yarn per feeder
Run resistance / stability	High (underlaps lock wales together)	Lower (jersey ladders, curls)
Pattern flexibility	Committed (beams + shog programme)	High, instant feeder/selection changes
Course rate	Very high (1000+ /min tricot)	Courses/min = feeders × rpm (circular)
Shaping	Essentially flat fabric only	Fully-fashioned and whole-garment possible

Glossary

Guide bar — bar carrying one yarn guide per needle space. **Overlap** — swing + shog that lays yarn into the hooks. **Underlap** — sideways shift behind the needles linking courses across wales. **Shog** — any lateral guide-bar shift, in needle spaces. **Chain notation** — numeric record of guide positions per course. **Rack** — 480 courses. **Run-in** — yarn fed per rack per bar, mm. **Tricot / Raschel** — the fine high-speed / versatile lace-technical warp machine families.

Key points

• Warp knitting = simultaneous loops + guide-bar underlaps; the underlap programme is the design.
• Pillar alone makes no fabric; opposed tricot bars make the standard ground; inlay adds stability without loops.
• Control variable is run-in (mm/rack) per bar, not a stitch cam. Verified tricot output: 29.5 m/h and 12.2 kg/h at 90 % for the stated machine, GSM ≈ 125.

Practice

1. Write the chain notation for a pillar bar and for one tricot bar from Figure 1, and mark the underlap in each.
2. A tricot machine runs 1500 courses/min at CPC 25/cm and 88 % efficiency. Find m/h.
3. Bar run-in is 1600 mm/rack; the machine makes 100 racks/h; ends = 3000; yarn 11 tex. Find that bar's kg/h (theoretical).
4. Why can a warp knit not be unravelled course by course like a weft knit? Answer from the underlap.
5. Choose warp or weft for: (a) run-resistant lingerie fabric in bulk, (b) a shaped sweater, (c) sports-shoe mesh with laid-in stability yarns. Justify each.

Answer key

1. Pillar: 1-0 / 0-1 repeating — the guide returns to the same wale; underlap = 0 spaces. Tricot bar: 1-0 / 1-2 repeating — between courses the guide stands one space over; underlap = 1 space, alternating direction.
2. Length = 1500 × 60 ÷ 25 ÷ 100 = 36.0 m/h; × 0.88 = **31.7 m/h**.
3. Per end per hour = 100 × 1600 mm = 160,000 mm = 160 m. Total = 3000 × 160 = 480,000 m = 480 km. Mass = 480 × 11 = 5280 g = **5.28 kg/h**.
4. Each loop is locked sideways by its underlap into the neighbouring wale and is overlapped by other bars' yarns; pulling one end does not release a course — the yarn is trapped in two directions, so the fabric resists laddering and unravelling.

5. (a) Warp (tricot) — run-resistance and bulk speed. (b) Weft (flat) — shaping is a weft capability. (c) Warp (raschel) — inlay bars lay stability yarns straight into a knitted ground at production speed.
